

Thomas Marconi

Software Engineer

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EDUCATION

Georgia Institute of Technology

M.S. Computer Science | In Progress, Part-Time

Atlanta, GA (Remote)

Expected Dec 2027

Florida State University

B.S. Computer Science

Tallahassee, FL

May 2023

EXPERIENCE

Tactical Air Support

Software Engineer 2

Software Engineer 1

Jacksonville, FL

Feb 2025 – Present

Feb 2024 – Feb 2025

- Led a 3-person team in delivering a real-time collaborative task platform adopted across 3 departments, replacing fragmented workflows and serving as the primary operational tool for engineering, IT, and aircraft integration teams.
- Replaced 3 enterprise vendor systems with full-stack solutions (Next.js, Node.js, FastAPI, PostgreSQL), resulting in \$850K+ in annual licensing savings.
- Enhanced security across 8+ microservices by eliminating static credentials and implementing runtime secrets injection via Managed Identity and Key Vault.
- Launched a retrieval-augmented generation (RAG) pipeline using LangGraph and Azure Semantic Search, reducing query time from 10 minutes to seconds and cutting QA deadline by 3 weeks.
- Unified Smartsheet and QT9 data sources into a FastAPI/MongoDB backend, creating a live operational dashboard and enabling 60% faster compliance reporting.
- Containerized 6+ applications using Docker and Podman, cutting deployment time by 30+ minutes and reducing bare-metal redeployment from hours to minutes across on-premises infrastructure.
- Implemented continuous integration (CI) pipelines using GitLab CE, accelerating the software release cycle by 35% and reducing manual errors.
- Spearheaded the implementation of automated backend testing frameworks using Jest, leading to a 40% reduction in bug reports post-deployment and improved code reliability.
- Introduced Agile from scratch as team grew from 1 to 3 engineers, establishing bi-weekly sprint cadence that replaced ad-hoc development and enabled consistent, scheduled releases for the first time.

PROJECTS

Project Desert | Next.js, Python (FastAPI), PostgreSQL, Docker

Jan 2026 – Present

- Enforced explicit API contracts via OpenAPI schema validation and auto-generated TypeScript clients, reducing integration failures by 50% across all API services.
- Integrated stateless JWT authentication with server-side refresh token rotation, securing 27 API endpoints from unauthorized requests.

CityScale | React, FastAPI, PCA | 3rd Place – AI for Good Hackathon

Oct 2025

- Spearheaded the design of housing risk indices across 25 U.S. cities using PCA, trained across 4,519 census tracts, achieving 66% explained variance across 6 principal components.
- Engineered a full stack geospatial affordability platform featuring a real-time, user-weighted PCA scoring engine and an LLM-powered chatbot to analyze 587,754 data points across 176 indicators, achieving top 3 ranking.

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, JavaScript, C++, Java, Bash

Backend: Node.js, FastAPI, WebSockets, REST APIs, PostgreSQL, Docker, Podman

Frontend: React, Next.js, Zustand, TanStack Query, Tailwind CSS

AI & Data: Vector Retrieval (ChromaDB, Azure Semantic Search), LLM Orchestration (LangGraph), Pandas, scikit-learn

Cloud & Infrastructure: Azure (Functions, Managed Identity, Key Vault, Entra ID), CI/CD, Linux

CERTIFICATIONS

Certified Scrum Master

Scrum Alliance

Jan 2025 – Present